

SUNDSVALL RASTA V, Sweden

WMO: 023540

Lat: 62.4306N

Lon: 17.3447E

Elev: 80

StdP: 100.37

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.9	-16.6	-21.5	0.6	-18.8	-19.0	0.7	-16.2	7.3	-1.3	6.5	-1.7	1.3	0	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	25.5	17.7	23.7	16.9	21.9	15.9	18.9	23.3	17.8	22.1	16.8	20.6	2.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	12.4	20.8	16.2	11.6	19.7	15.1	10.9	18.5	53.9	23.2	50.4	22.3	47.4	20.6	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.3	5.5	4.9	-21.0	28.9	3.7	1.8	-23.7	30.1	-25.8	31.2	-27.9	32.2	-30.7	33.5		
			-20.9	20.2	3.7	1.1	-23.5	21.0	-25.7	21.6	-27.8	22.2	-30.5	23.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.1	-5.2	-4.5	-0.5	3.8	8.8	13.3	16.7	15.2	10.7	4.9	0.3	-3.3
	DBStd	8.52	5.52	5.62	4.27	3.29	3.42	3.01	2.70	2.67	3.12	3.57	4.18	5.56
	HDD10.0	2356	471	407	325	189	64	5	0	1	28	160	292	414
	HDD18.3	4866	729	640	583	437	294	154	67	103	228	416	541	672
	CDD10.0	555	0	0	0	2	28	102	208	162	50	2	0	0
	CDD18.3	23	0	0	0	0	0	2	16	6	0	0	0	0
	CDH23.3	197	0	0	0	0	6	41	120	31	0	0	0	0
	CDH26.7	28	0	0	0	0	0	6	19	3	0	0	0	0
Wind	WSAvg	2.2	2.2	2.0	2.3	2.2	2.3	2.4	2.1	2.0	2.0	2.0	2.1	2.1
Precipitation	PrecAvg	747	54	40	46	49	42	55	77	86	79	75	82	62
	PrecMax	1005	141	120	124	125	124	154	203	253	258	204	220	235
	PrecMin	476	0	4	8	4	5	5	8	15	8	13	17	5
	PrecStd	124	32	26	29	34	23	35	45	55	49	48	51	40
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.0	7.5	12.1	18.4	23.7	27.2	28.8	26.4	21.4	14.8	10.7	7.9
		MCWB	3.7	4.5	7.3	11.4	15.7	17.8	19.0	17.1	15.2	10.6	8.3	5.7
	2%	DB	4.4	5.6	9.3	15.0	19.8	24.1	26.4	23.9	18.9	13.0	8.9	6.1
		MCWB	2.6	3.2	5.2	9.1	13.0	16.2	18.3	17.1	14.0	9.9	7.0	4.1
	5%	DB	3.1	4.0	7.3	12.2	17.6	21.7	24.4	22.1	17.2	11.4	6.9	4.8
		MCWB	1.6	2.0	3.9	7.3	11.4	14.7	17.5	16.4	13.0	9.3	5.5	3.2
	10%	DB	1.7	2.5	5.6	10.3	15.5	19.6	22.6	20.3	15.7	10.2	5.5	3.5
		MCWB	0.5	0.9	2.8	5.8	10.2	13.5	16.7	15.7	12.3	8.4	4.5	2.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.0	4.7	7.4	11.7	16.1	18.5	20.7	20.1	16.0	12.1	8.9	5.8
		MCDB	5.5	7.2	11.8	17.1	22.5	25.5	26.6	24.1	20.2	13.8	10.4	7.6
	2%	WB	2.9	3.5	5.5	9.6	13.5	16.8	19.2	18.4	14.7	10.6	7.1	4.4
		MCDB	4.2	5.3	8.9	14.3	18.9	22.7	24.0	21.5	17.5	11.9	8.6	5.8
	5%	WB	1.8	2.2	4.2	7.7	11.9	15.5	18.4	17.2	13.9	9.6	5.8	3.4
		MCDB	2.9	3.8	6.9	11.6	16.8	20.4	23.1	20.2	16.1	11.3	6.7	4.7
	10%	WB	0.6	1.0	3.0	6.2	10.5	14.2	17.4	16.3	13.0	8.5	4.7	2.2
		MCDB	1.5	2.3	5.5	9.6	14.8	18.8	21.7	19.3	15.0	10.1	5.4	3.3
Mean Daily Temperature Range	5% DB	MDBR	5.2	6.0	7.2	8.5	9.1	9.2	8.7	8.1	7.7	6.2	4.2	5.0
		MCDBR	4.9	6.4	9.1	12.8	12.8	13.3	11.5	10.7	9.6	8.0	5.6	6.2
		MCWBR	4.2	5.1	6.1	7.6	7.0	6.6	5.5	5.2	5.6	5.5	4.5	5.2
	5% WB	MCDBR	4.7	6.0	8.9	12.3	12.1	12.2	10.4	8.7	8.3	7.5	5.2	5.9
		MCWBR	4.1	4.9	6.1	7.7	7.0	6.4	5.5	4.8	5.4	5.5	4.5	5.1
Clear-Sky Solar Irradiance	taub	0.237	0.293	0.316	0.352	0.354	0.349	0.366	0.358	0.325	0.311	0.254	0.345	
	taud	2.113	2.252	2.351	2.378	2.437	2.485	2.456	2.473	2.548	2.449	2.180	2.624	
	Ebn at Noon	438	661	795	835	865	876	848	824	790	649	409	270	
	Edh at Noon	32	61	80	96	99	96	97	87	67	51	30	20	
All-Sky Solar Radiation	RadAvg	0.20	0.84	2.32	3.77	4.80	5.55	5.07	3.85	2.42	1.06	0.30	0.11	
	RadStd	0.03	0.14	0.26	0.32	0.49	0.35	0.54	0.32	0.30	0.19	0.06	0.01	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page